

6-2-1: Examining Single Sign-on

At the end of this episode, I will be able to:

1. Identify and explain the importance and components of single sign-on or SSO technologies, given a scenario.

Learner Objective: *Identify and explain the importance and components of single sign-on or SSO technologies, given a scenario*

Description: In this episode, the learner will examine single sign-on or SSO technologies and the importance and components of security. We will explore identity federation, identity providers or IdPs, SAML, OAuth, and OpenID concepts.

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- - What is single sign-on (SSO) and how is it beneficial?
 - An authentication process that uses technologies enabling users to access multiple applications, services and resources using a single set of login credentials.
 - What are some of the benefits of (SSO)?
 - Eliminating multiple usernames and passwords to be entered.
 - Simplifying the user experience
 - Centralized access control to multiple services
 - Increased security with MFA, conditional access (attribute-based access control)
 - What is identity federation?
 - A mechanism allowing users to access multiple systems, applications or resources using a single set of login credentials, reducing the burden of authentication and improving user experience.
 - What is an identity provider (IdP)
 - A trusted entity that authenticates users and provides identity information to a service provider
 - Can be internal IdPs such as ADDS or LDAP server
 - Can be external IdPs such as large service providers like Google, Facebook and Microsoft
 - Is there a benefit to using IdPs?
 - We use them daily on the web, applications, the cloud and on-premise.
 - Centralized management
 - Familiarity (better user experience)
 - Enabling seamless, secure access to various resources with a single login.
 - External IdPs provide large user identity databases for a fee
 - What is SAML?
 - A standardized, XML-based way to communicate authentication data once, then reuse that information across multiple resources.
 - What is OAuth?
 - An authorization *framework* (as opposed to a protocol) in SSO implementations enabling IdPs to limit access permissions, presented to a Service Provider (SP) on behalf of a user, allowing seamless access to protected resources without sharing the user's login credentials.
 - Service provider = commonly a web application
 - Separates out the authorization service and API server (API server only receives OAuth tokens)
 - What is OpenID?
 - Adds an identity layer to OAuth2.0, providing information about the user.
 - Enables client login sessions
 - What is Kerberos?
 - A network authentication protocol enabling secure authentication between clients and servers within a network, facilitating single sign-on and data encryption for enhanced security.

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- Additional Resources:
 - If applicable